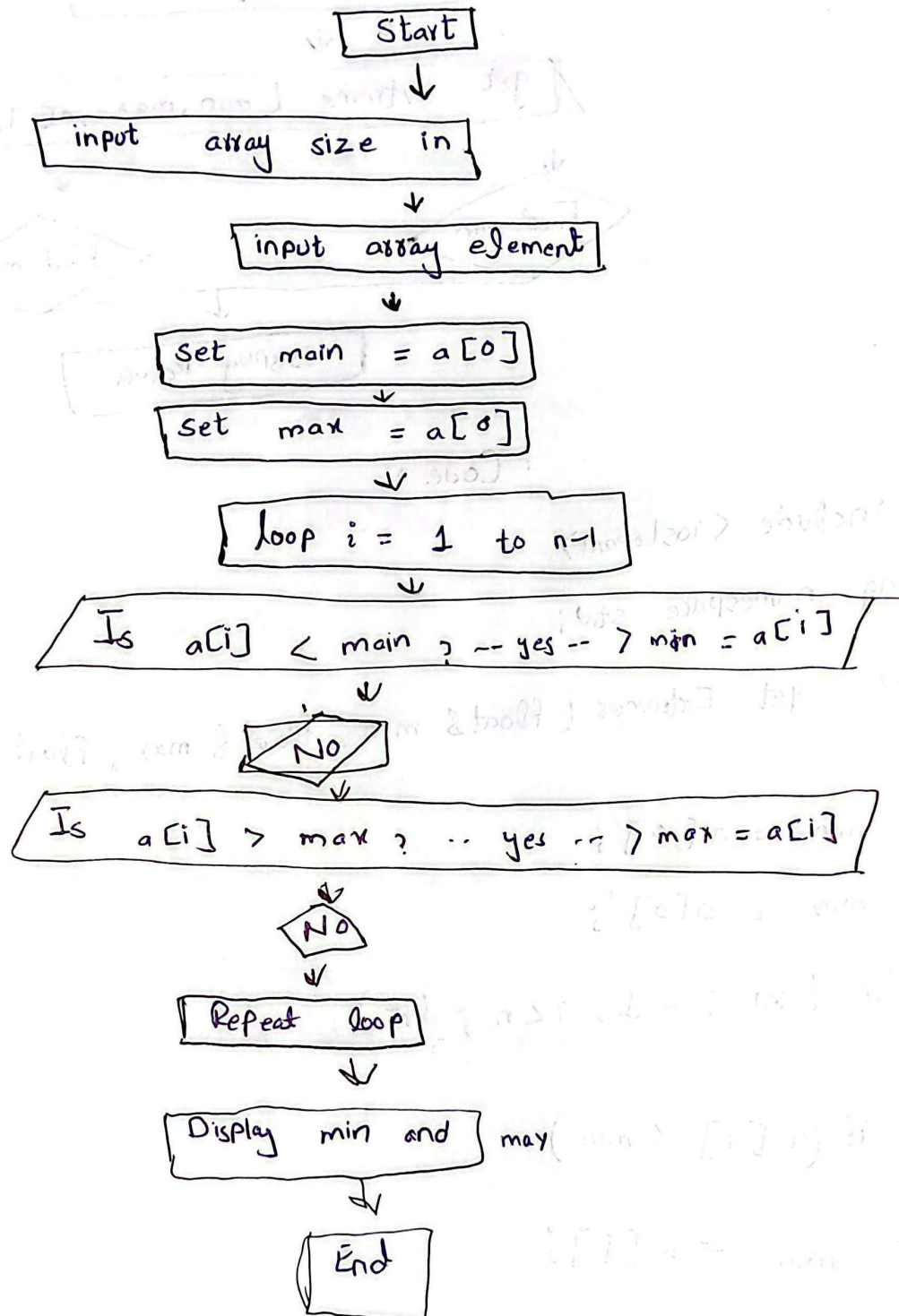
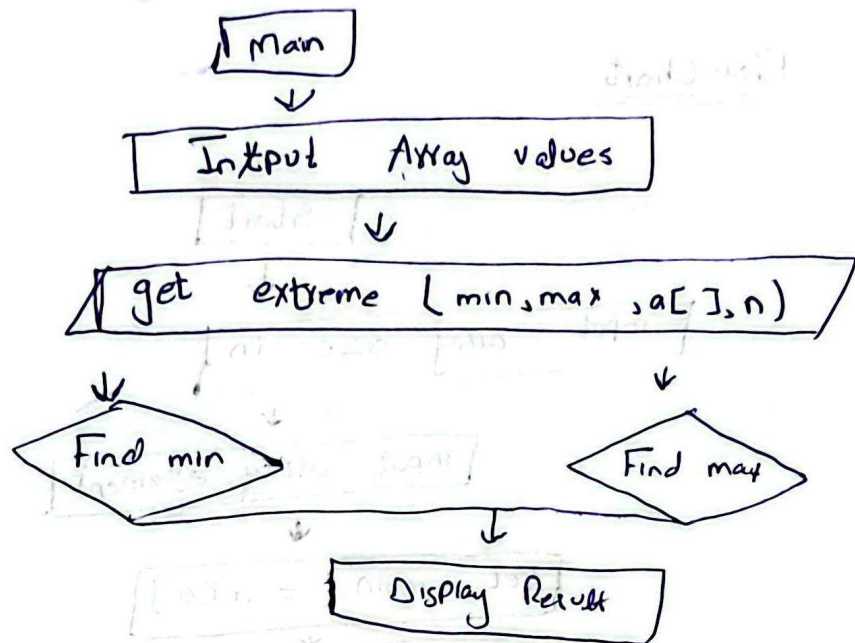


Question 1:

Flow Chart



Block Diagram



Code :

```
#include <iostream>
using namespace std;

void get Extremes ( float & min , float & max , float a[] , int n )
{
    min = a[0];
    max = a[0];

    for ( int i = 1; i < n; i++ )
    {
        if ( a[i] < min )
        {
            min = a[i];
        }
        if ( a[i] > max )
        {
            max = a[i];
        }
    }
}
```

```
}  
}  
}
```

```
int main ()
```

```
{  
    int n;
```

```
    cout << "Enter size of Array":;
```

```
    cin >> n;
```

```
    float arr [n];
```

```
    cout << "Enter array elements :\n":;
```

```
    for (int i = 0; i < n; i++)
```

```
    {  
        cin >> arr [i];
```

```
    }
```

```
    float minimum , maximum;
```

```
    getExtremes (min , max , arr , n);
```

```
    cout << "Minimum value = " << min << endl;
```

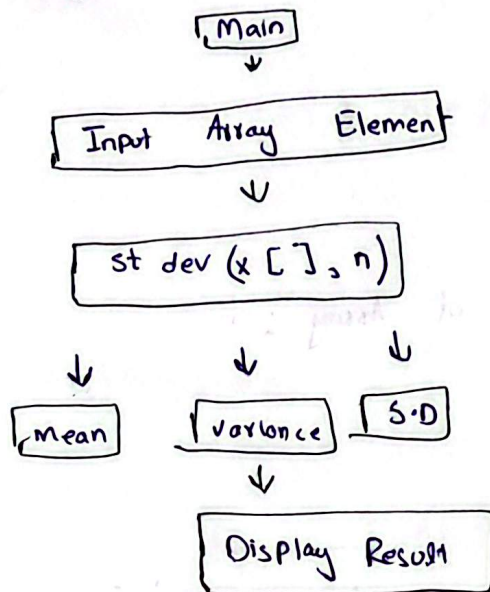
```
    cout << "Maximum value = " << max << endl;
```

```
    return 0;
```

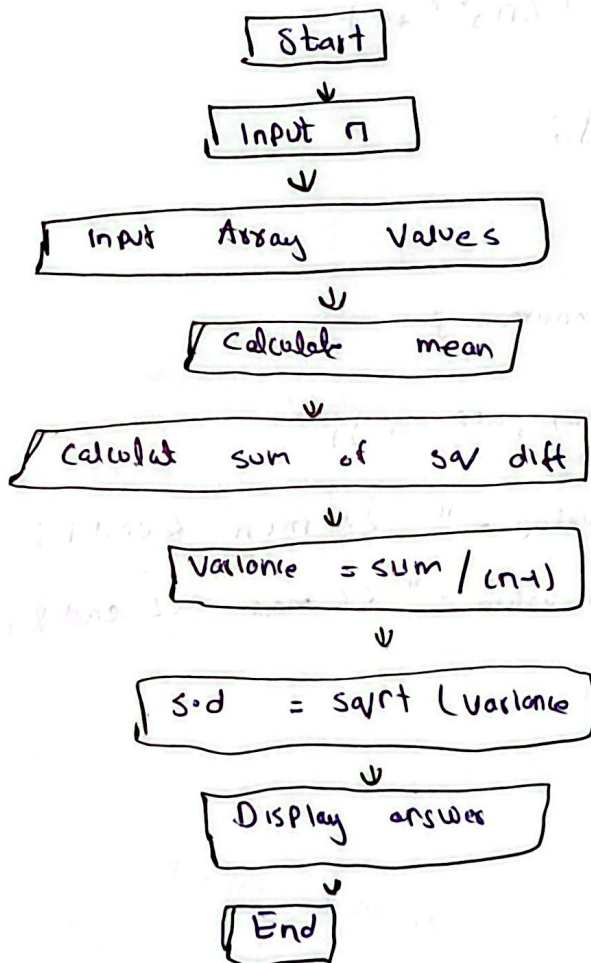
```
}
```

Question 2

Block Diagram



Flowchart



```
# include <iostream>
```

```
# include <cmath>
```

```
using namespace std;
```

```
double stdev (double x[], int n)
```

```
{
```

```
    double sum = 0;
```

```
    double mean;
```

```
    double variance = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        sum += x[i];
```

```
    }
```

```
    mean = sum / n;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        variance = variance + (x[i] - mean) * (x[i] - mean);
```

```
    return sqrt (variance / (n - 1));
```

```
}
```

```
int main ()
```

```
{
```

```
    int n;
```

```
    cout << "Enter number of elements:";
```

```
    cin >> n;
```

```
    double arr[n];
```

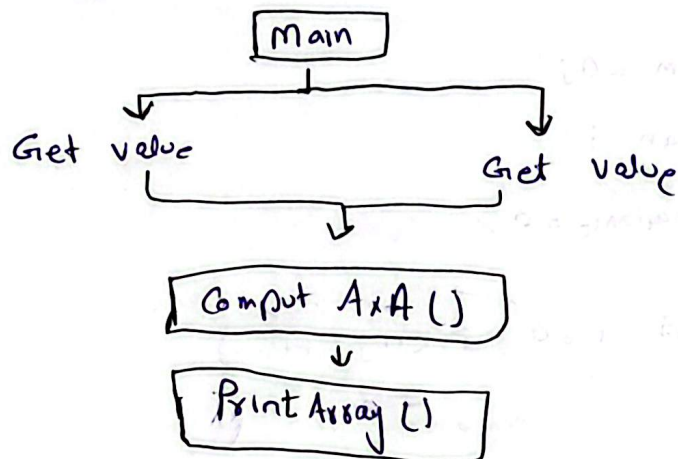
```

    cout << "standard Deviation" << endl;
    return 0;
}

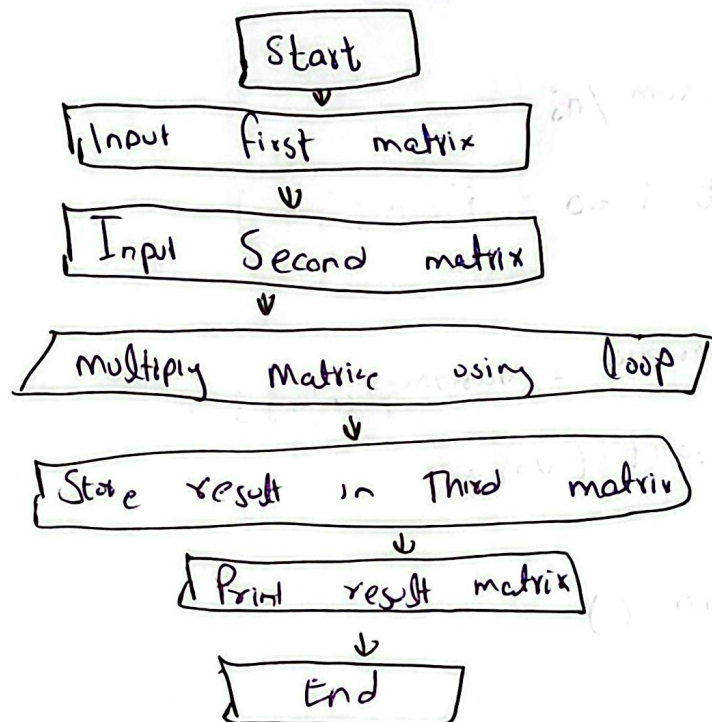
```

Question 3.

Block Diagram



Flow Chart



Question 4

Block Diagram

```
cout << "std" = " << std::dev(arr, n);  
return 0;  
}
```

Code: 3

```
#include <iostream>  
using namespace std;  
void Get value (int arr[3][3])  
{  
    for (int i = 0; i < 3; i++)  
    {  
        for (int j = 0; j < 3; j++)  
        {  
            cin >> arr[i][j];  
        }  
    }  
}  
void compute AXA (int a[3][3], int b[3][3], int result[3][3])  
{  
    for (int i = 0; i < 3; i++)  
    {  
        for (int j = 0; j < 3; j++)  
        {  
            result[i][j] = 0;  
            for (int k = 0; k < 3; k++)  
            {  
                result[i][j] += a[i][k] * b[k][j];  
            }  
        }  
    }  
}
```

```

    }
}

void Print Array (int arr[3][3])
{
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 3; j++)
        {
            cout << arr[i][j] << " ";
        }
        cout << endl;
    }
}

```

```

int main()

```

```

{
    int A[3][3], B[3][3], C[3][3];

```

```

    cout << "Enter first matrix : \n";

```

```

    Get value (A);

```

```

    cout << "Enter second matrix : \n";

```

```

    Get value (B);

```

```

cout << "Enter

```

```

    compute A x A (A, B, C);

```

```

    cout << "Result matrix : \n";

```

```

    Print Array (C);

```

```

    return 0;

```

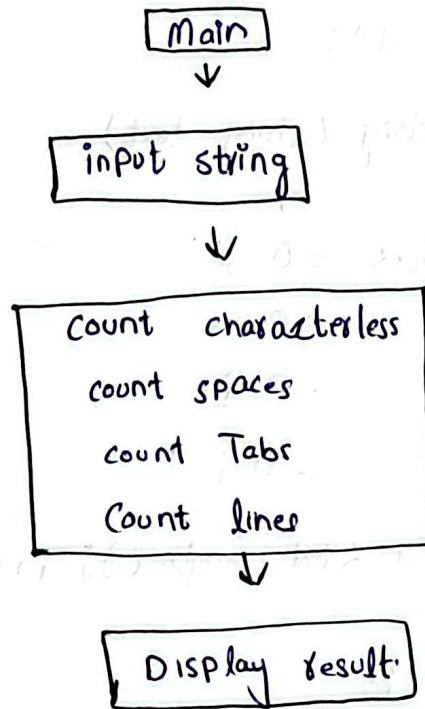
```

}

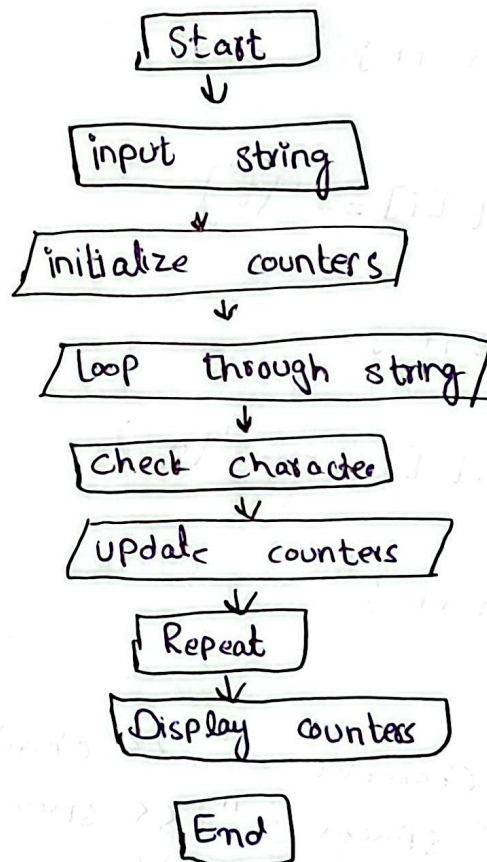
```

Question 4

Block Diagram



Flow Chart :



Code:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
void analyze string (string text)
```

```
{
```

```
    int characters = 0;
```

```
    int spaces = 0;
```

```
    int tabs = 0;
```

```
    int lines = 0;
```

```
    for (int i = 0; i < text.length(); i++)
```

```
    {
```

```
        characters++;
```

```
        if (text[i] == ' ')
```

```
        {
```

```
            spaces++;
```

```
        }
```

```
        if (text[i] == '\t')
```

```
        {
```

```
            tabs++;
```

```
        }
```

```
        if (text[i] == '\n')
```

```
        {
```

```
            lines++;
```

```
        }
```

```
    }
```

```
    cout << " Characters = " << characters << endl;
```

```
    cout << " Spaces = " << spaces << endl;
```

```
    cout << " Tabs = " << tabs << endl;
```

```
    cout << " Lines = " << lines << endl;
```

```
}
```

```

int main ()
{

```

```

    string text;

```

```

    cout << "Enter text:";

```

```

    get_line (cin, text);

```

```

    analyze string (text);

```

```

    return 0;

```

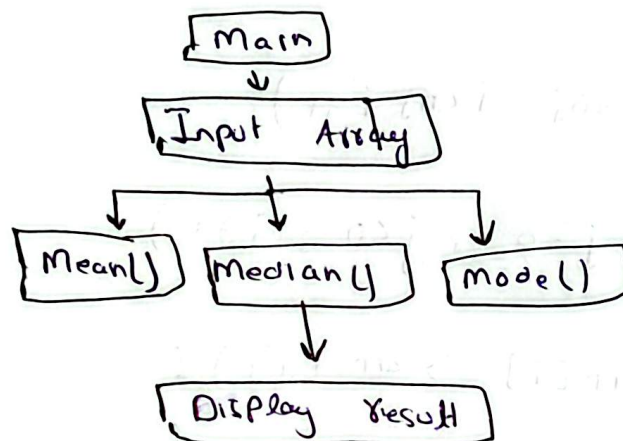
```

}

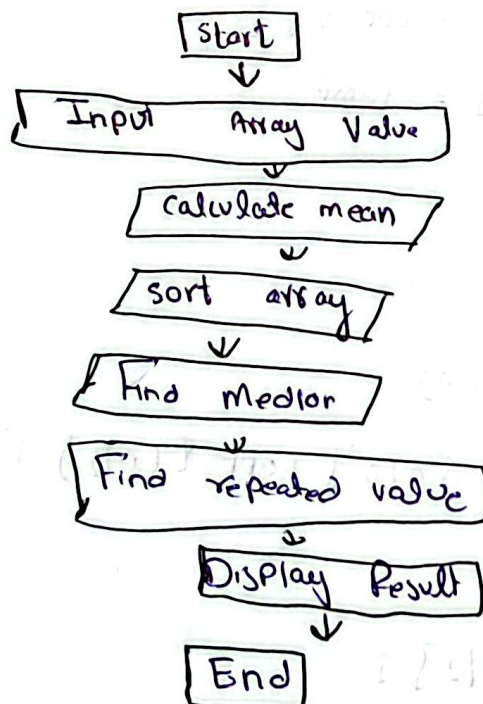
```

Question 5

Block Diagram



Flow Chart:



Code:

```
#include <iostream>
```

```
using namespace std;
```

```
float Mean (int arr[], int n)
```

```
{  
    int sum = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {  
        sum += arr[i];
```

```
    }  
    return (float) sum / n;
```

```
}  
float median (int arr[], int n)
```

```
{  
    for (int i = 0; i < n; i++)
```

```
    {  
        for (int j = 0; j < n; j++)
```

```
        {  
            if (arr[i] > arr[j])
```

```
            {  
                int temp = arr[i];  
                arr[i] = arr[j];  
                arr[j] = temp;
```

```
            }
```

```
        }
```

```
    }
```

```
    if (n % 2 == 0)
```

```
    {  
        return (arr[n/2] + arr[(n/2) - 1]) / 2.0;
```

```
    }
```

```
    else
```

```
        return arr[n/2];
```

```
    }
```

```
}
```

```
int mode (int arr[], int n)
```

```
{
```

```
    int mode = arr[0];
```

```
    int maxCount = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        int count = 0;
```

```
        for (int j = 0; j < n; j++)
```

```
        {
```

```
            if (arr[j] == arr[i])
```

```
            {
```

```
                count++;
```

```
            }
```

```
        }
```

```
        if (count > maxCount)
```

```
        {
```

```
            maxCount = count;
```

```
            mode = arr[i];
```

```
        }
```

```
    }
```

```
    return mode;
```

```
}
```

```
int main ()
```

```
{
```

```
    int n;
```

```
    cout << "Enter size of array: ";
```

```
    cin >> n;
```

```
    int arr[n];
```

```
    cout << "Enter element: \n";
```

```

for (int i = 0; i < n; i++)
{
    cin >> arr[i];
}

cout << "mean = " << mean(arr, n) << endl;
cout << "Median = " << median(arr, n) << endl;
cout << "Mode = " << mode(arr, n) << endl;

return 0;
}

```